Lean cutoff for the printed Joux–Vitse empty example

Lean proves a proper homogeneous ideal I in F101[x,y,h] contains every monomial of total degree at least four. Its dehomogenized ideal at h=1 is the unit ideal: there is no affine solution in any characteristic101 field. The three literal quadratics are from the first $F101^3$ worked example on page7 of Joux–Vitse's author manuscript. Here x,y are the elementary symmetric coordinates, h is the homogenizing variable. No generic regularity premise is assumed. Whole gate: 39 declarations, zero failures and zero trusted-native exceptions. [pcc:unverified]

Equation	x^2	xy	y^2	xh	yh	h^2
f0	100	32	1	55	17	81
f1	29	34	0	71	83	16
f2	59	27	0	31	48	32

Universal characteristic101 identities give normalized quadratics g_y, g_{xy}, g_x and exact ideal-membership certificates for

$$c_y = yh^2 + 82h^3, \qquad c_x = xh^2 + 66h^3,$$

$$h^4 = 77(h^2 g_x - x c_x - 30 h c_x - 64 h c_y).$$

These are kernel-checked polynomial identities. No assumed Gröbner basis, membership oracle or native execution result is used. [pcc:unverified]

All nine cubic normal forms reduce to specified scalars times h^3. The checked quartic relation and their products make every degree-four monomial vanish in the actual quotient. Properness is proved by evaluation at the origin before obtaining its characteristic101 structure. Explicit exponent factorization extends ideal membership to every larger total degree. [pcc:unverified]

The closed declaration, in namespace JVMacaulayAudit, is `literal_jv_macaulay_upper_certificate`. It proves the actual affine unit ideal, homogeneous properness and all higher monomial membership. Only propext, Classical.choice and Quot.sound occur. Lean4.35.0 / Mathlib56a00585 are pinned; no holes, custom axioms, native_decide, waivers or raised elaboration limits. Failed attempts and complete gate receipts are retained losslessly. [pcc:unverified]

This proves an algebraic upper cutoff of four, not its minimality, exact Hilbert function or quotient length8. Native execution, complete internal F4degrees and formal $F101^3$ curve/coefficient-descent identification remain separate. The frozen native study returns GBdegree4 and then a degree5 selected-pair round with zero new polynomials/two zero reductions. That log alone is not promoted to an Assumption2 refutation. Generic regularity, larger m,n cases and the general computation-degree claim remain open. [pcc:unverified]

Source: Joux–Vitse DLFq5.pdf pp7/9–10, SHA256 6f3268a0ef8c50b5… (full digest in source-review.json); exact read scope/source pins accompany this report. KB unavailable (MCP absent/wrapper127); bounded primary/repository/Mathlib fallback. No new native experiment, novelty claim, paid compute or agent storage upload. Exact PCC/source receipts accompany the report; signed standing is distinct from local kernel replay. [pcc:unverified]